How many patients should Meridian contact?

1,368 whose expected savings beat the cost of contact, plus a 1,100-patient learning band.

The bottom line

1,368

patients above the 1.45% break-even risk: contact them

1,100

more just below it, randomized 50/50 to learn whether outreach works there too

1,644

contacts in total; 824 held out to measure the effect

What this means

- The 1,094 tier-1 contacts are expected to prevent about 0.9 surgeries this wave: $6.8k net of contact costs.
- The learning band costs about $0.6k in expected value: the price of checking the cut-off is in the right place.
- The number depends most on how much enrollment reduces surgery, which the data can't measure.

The pilot is built around that.

The rule: contact while savings beat cost

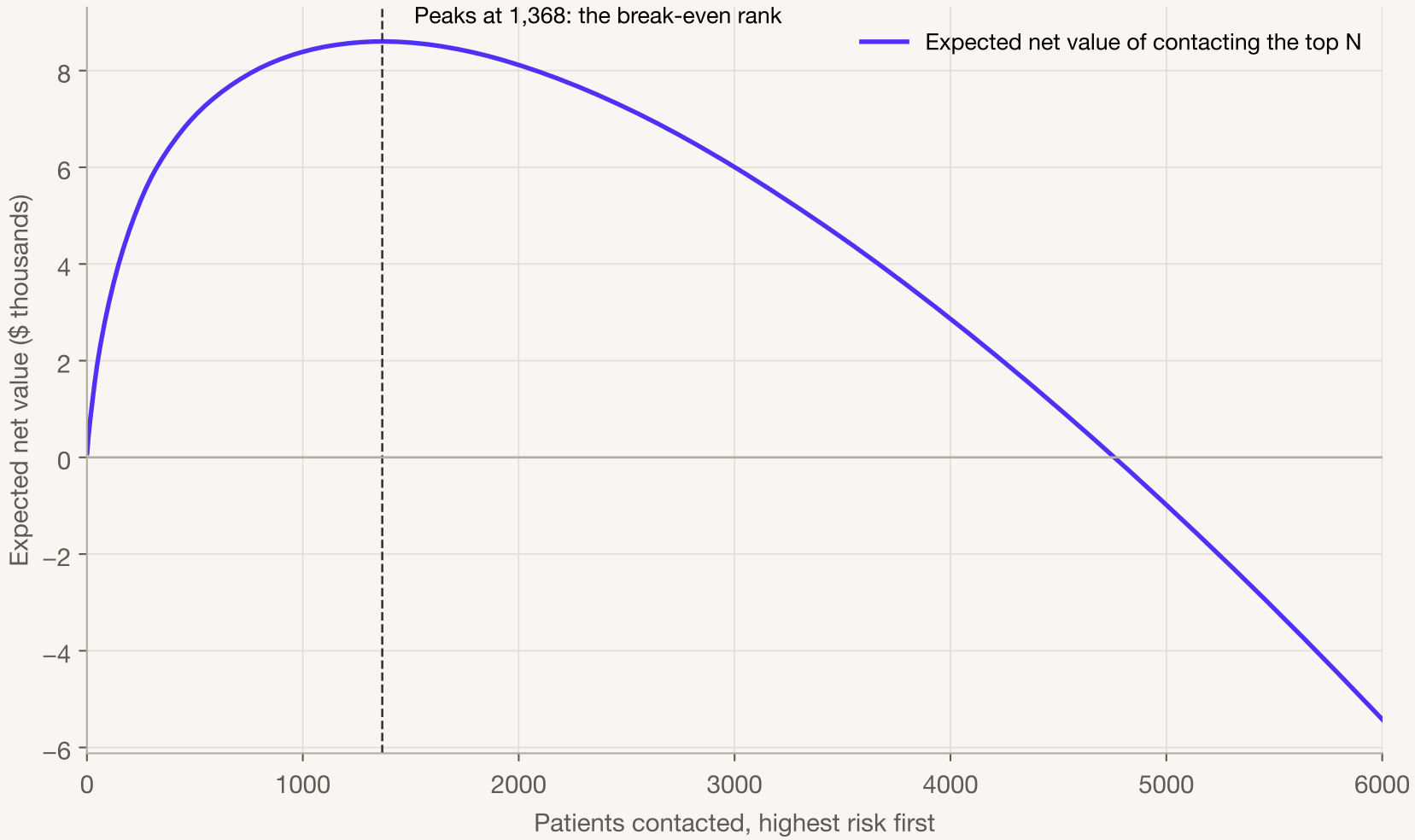
P(surgery) × enrollment lift × surgery reduction × surgery cost > contact cost

Input	Value	Source
Enrollment lift	11.5%	Estimated from past campaigns (doubly robust, 89% interval 10.6–12.4%)
Surgery reduction per enrollment	30%	Literature assumption: the data can't test it
Surgery cost	$20,000	Medicare 90-day episode, fusion/decompression blend
Contact cost	$10	Vi call: amortized cost per contacted patient ($5–$10)
Break-even risk	1.45%	$10 ÷ (0.115 × 0.3 × $20,000)

P(surgery) comes from the Surgery Risk Model, calibrated out-of-time (2.16% predicted vs 2.16% observed).

HOW FAR DOWN THE LIST

The list is worth most at 1,368 patients



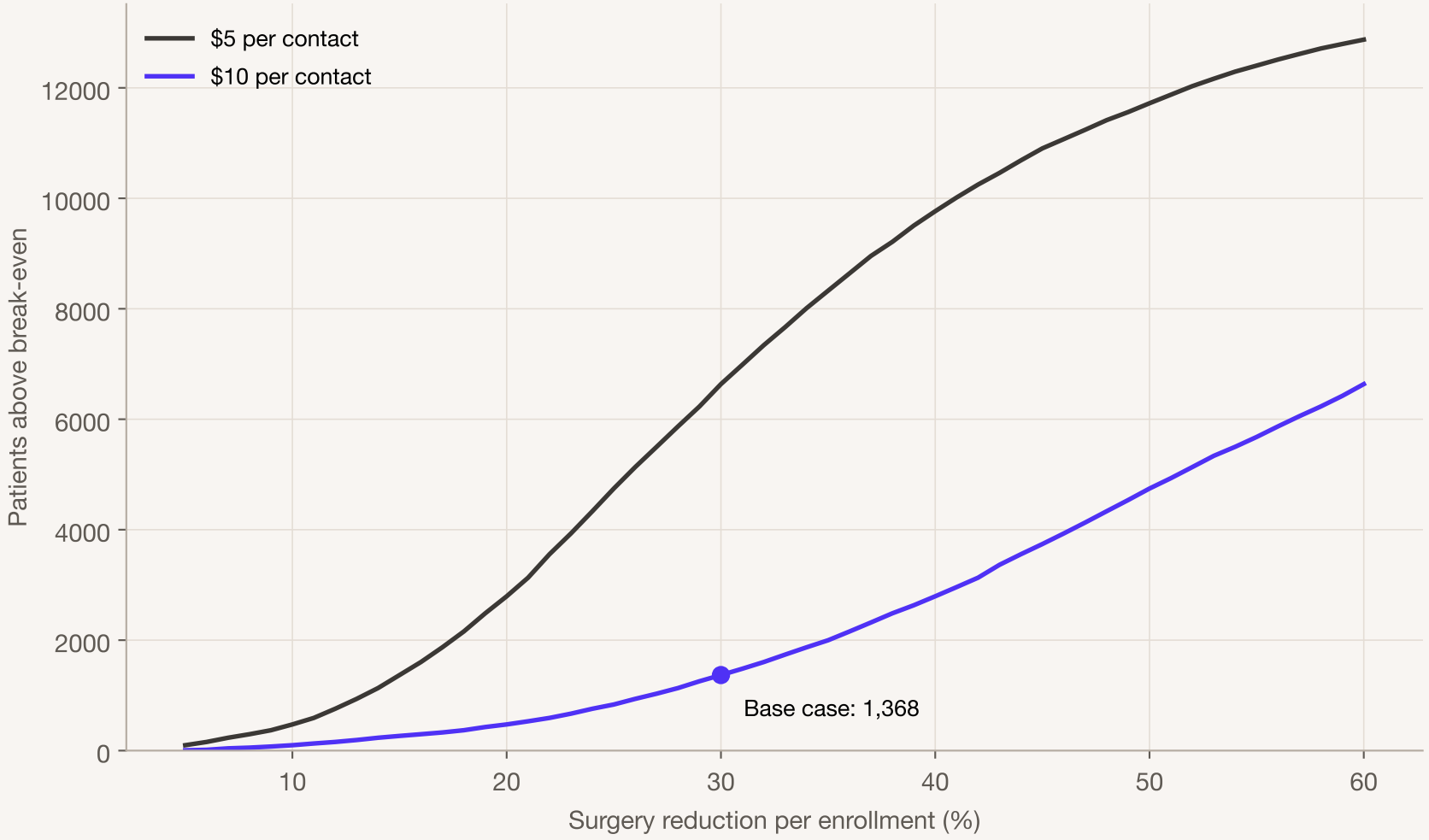
Each contact adds its expected savings minus its cost.

Above 1.45% risk, contacts pay for themselves; below it, each one subtracts value.

Peak: $8.6k expected net value, if every listed patient were contacted.

HOW FAR DOWN THE LIST

One assumption moves the answer most



Patients above break-even as the surgery reduction varies:

10% reduction: 98

50% reduction: 4,745

Halving the contact cost to $5 lifts the base case to 6,636.

The reduction is a literature assumption; the data can't test it.

HOW FAR DOWN THE LIST

Every assumption at its low and high value

Assumption	Value	Break-even risk	Patients to contact
Base case		1.45%	1,368
Surgery cost	$10,000	2.90%	266
Surgery cost	$40,000	0.72%	6,636
Contact cost	$5	0.72%	6,636
Surgery reduction	10%	4.35%	98
Surgery reduction	50%	0.87%	4,745
Enrollment lift	10.6%	1.57%	1,096
Enrollment lift	12.4%	1.34%	1,654

Read across a row to see what changes if one input is wrong.

The list ranges from 98 to 6,636 patients.

The rule stays the same; only the cut-off moves. Re-running the scoring step with a new assumption updates the list.

HOW FAR DOWN THE LIST

Tiers: contact, and learn as we go

Tier	Who	Contact / hold out	Contacted	Held out	Weight mgmt	Prior consult
1	Above break-even (1,368)	80% / 20%	1,094	274	409	340
2	Learning band (1,100)	50% / 50%	550	550	310	10
—	Below rank 2,468	Not contacted this wave				

- Randomized within tier and program with a fixed seed. Held-out patients get usual care, and Meridian's own campaigns must skip them for 12 months.
- Primary outcome: enrollment within 90 days. Detectable lift: 5.2% pooled, 8.2% in tier 1 alone, against 11.5% measured historically.
- Surgery can't be measured this way: a 0.3-point drop needs about 34,000 patients per arm.

What Meridian needs to confirm

Capacity

The list asks for 1,644 contacts this wave. Past waves reached 2,189–3,183 patients. If Meridian can do fewer, shrink the learning band first, then tier 1 from the bottom.

Contact cost

$10 per contacted patient is the conservative end of $5–$10. At $5, the list grows to 6,636.

Surgery cost

$20,000 is a published Medicare blend. Meridian's contracted rates would replace it: at $10,000 the list is 266.

Contact 1,368, learn from 1,100 more

- Contact everyone above 1.45% risk: 1,368 patients, 1,094 after the holdout.
- Randomize the next 1,100 50/50 to test whether the cut-off is in the right place.
- Re-run the list each wave with updated assumptions; the rule stays, the cut-off moves.